Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/23/2014
PI Name:	Clark, Noel A. Ph.D.		
Project Title:	Structure and Dynamics of Freely Suspended Liquid Crystals		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80309-0001	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/12/2007	**End Date:**	04/22/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/22/2013 in July 2009, per NSSC information; original end date was 4/11/2010 (Ed., 5/29/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Maclennan, Joseph (University of Colorado) Glaser, Matthew (University of Colorado)		
Grant/Contract No.:	NNX07AE48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ED. NOTE: Continuation of "Structure and Dynamics of Freely Suspended Liquid Crystals"; grant number NNC04GA50G. We pursue ground-based studies of freely suspended liquid crystal films with an eye toward µg experiments, as well as atomistic computer simulation of liquid crystal phases, probing the molecular origins of observed film behavior. Ground-based studies focus on tethered smectic bubbles on which islands (isolated pancake-shaped thicker regions) can be generated and observed. The islands form a novel interacting particle array confined to the two-dimensional surface of the bubble. We observe islands (using reflected light microscopy) and manipulate them (using optical tweezers) and carry out computer simulation of the interaction of islands on contact, in an effort to understand their stability against aggregation. Our work is directed toward developing a deeper understanding of condensed phases under conditions of reduced dimensionality. Films are locally optically anisotropic so that the pattern of molecular orientation within the film plane can be effectively studied by depolarized reflected light microscopy are also being studied. In particular, topological defect motion can be observed and the simplest liquid crystal defect dynamical process, the attraction and annihilation of oppositely charged, unit-strength defects (vortex-antivortex pairs), probed. Experiments to date reveal significant shortcomings in our understanding of this process. Application of in-plane electric fields to tilted smectic films reveals a host of interesting field-induced structures and transitions, and leads to novel methods for probing film structure. For example, by comparing the response of even layer numbered and odd layer numbered films, basic information about the nature of the polar ordering (ferroelectric or antiferroelectric) of the smectic can be obtained. Our recent discovery of the spontaneous formation of chiral domains in fluid smectic phases of achiral banana-shaped molecules (the first observation of spontaneous chiral domain formation in a fluid since Pasteur's discovery in crystals 150 years ago), based on observation of free film response to applied electric field, will be pursued, as will the recent discovery of freely suspended filaments of smectics comprised of banana-shaped molecules. Smectic liquid crystals are phases of rod shaped molecules organized into one dimensionally (1D) periodic arrays of layers, each layer being a two dimensional (2D) liquid between one and two molecular lengths thick. This inherent fluid-layer structure and the low vapor pressure of smectic liquid crystals enable the long term stabilization of freely suspended, single component, fluid films quantized in integral numbers of layers in thickness, down to a single molecular layer (30Å). These ultrathin freely suspended liquid crystal films are structures of fundamental interest in condensed matter and fluid physics. They are the thinnest known stable condensed phase fluid structures and have the largest surface-to-volume ratio of any stable fluid preparation, making them ideal for the study of the effects of reduced dimensionality on phase behavior and on fluctuation and interface phenomena. Their low vapor pressure and quantized thickness enable the effective use of microgravity to extend the study of basic capillary phenomena to ultrathin fluid films.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research Impact on America/Earth Benefits: fundamental studies of fluid dynamics and self-organization in two dimensions.
Task Progress:	(Ed. note Dec 2014: Compiled from PI's final report) Smectic liquid crystals composed of fluid layers can be made to form freely suspended, single component, fluid films as thin as a single molecular layer (3 nm). These films are of fundamental interest in condensed matter physics. They are the thinnest known stable fluid structures and have the largest surface-to-volume ratio of any fluid preparation, making them ideal for studies of two dimensional phase transitions, fluid flow, and fluctuation and interface phenomena. They offer unique opportunities to effectively use microgravity to extend the study of basic capillary phenomena to ultrathin fluid films. The smectic layering forces films to be an integral number of layers thick, quantizing their thickness in layer units and forcing a film of a particular number of layers N to be physically homogeneous with respect to its layer structure over its entire area. This inherent fluid-layer structure and the low vapor pressure of smectics enable, in a straightforward way, the stabilization of freely suspended, single component, layered fluid films for detailed study. The main objective of the research supported by this Grant is to advance the understanding of fluid physics by exploiting the unique characteristics of freely suspended liquid crystal films in a microgravity environment. These experiments focus on using a novel collective system of interacting, one dimensional interfaces in two dimensional fluids for the study of fluid physics in two dimensions and the capillarity of ultrathin fluid films. We are pursuing the definition of experiments on island generation, observation (using reflected light microscopy), and manipulation (using optical tweezers) for the µg environment. This period of time has seen the following progress: A. Crossover between 2D and 3D Fluid Dynamics in the Diffusion of Islands in Ultrathin Freely Suspended Smectic Films B. Two-Dimensional Microrheology of Freely Suspended Liquid Crystal Films C. Surface Energetics of a Freely Suspended Fluid Molecular Monolayer D. Novel type of polar behavior of bulk paralelectric SmC phase in thin films: transitions between para-, ferro- and antiferroelectric states E. Hydrodynamics of smectic islands F. Two-dimensional Domain Coalescence Driven by Line Tension
Bibliography Type:	Description: (Last Updated: 12/04/2024)
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Articles in Peer-reviewed Journals	Chattham N, Korblova E, Shao R, Walba DM, Maclennan JE, Clark NA. "de Gennes' triclinic smectics – not so far-fetched after all." Liquid Crystals. 2009 Oct-Nov;36(10-11):1309-17. http://dx.doi.org/10.1080/02678290903306458 , Oct-2009
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